Quantitative methods in Biology

LS1102

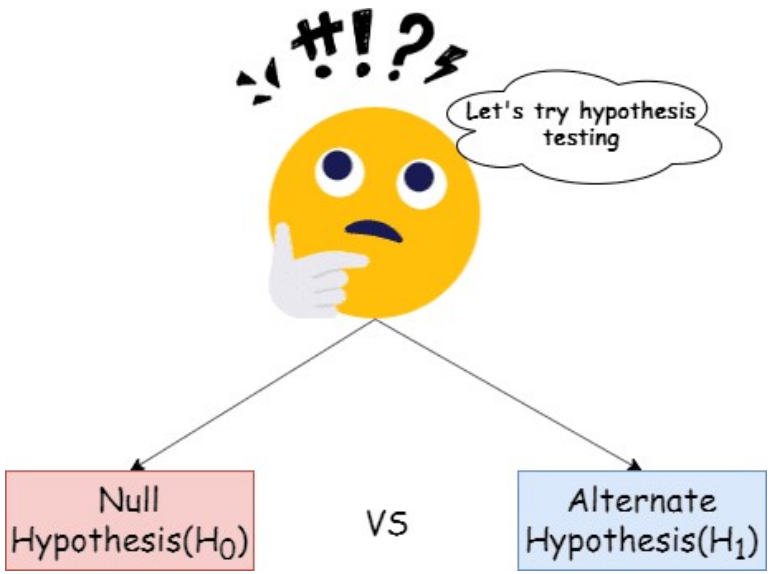
Specify the biological question you are asking.

Putting the question in the form of a biological null hypothesis and alternate hypothesis.

The null hypothesis is a statement that you want to test.

In general, the null hypothesis is that things are the same as each other, or the same as a theoretical expectation.

The alternative hypothesis is that things are different from each other, or different from a theoretical expectation.



NULL HYPOTHESIS

THE NULL HYPOTHESIS ASSUMES THERE IS NO DIFFERENCE BETWEEN TWO GROUPS.

NULL HYPOTHESIS
H_0

LIGHT COLOR HAS NO EFFECT ON PLANT GROWTH

ALTERNATIVE HYPOTHESIS
H_A

LIGHT COLOR AFFECTS PLANT GROWTH

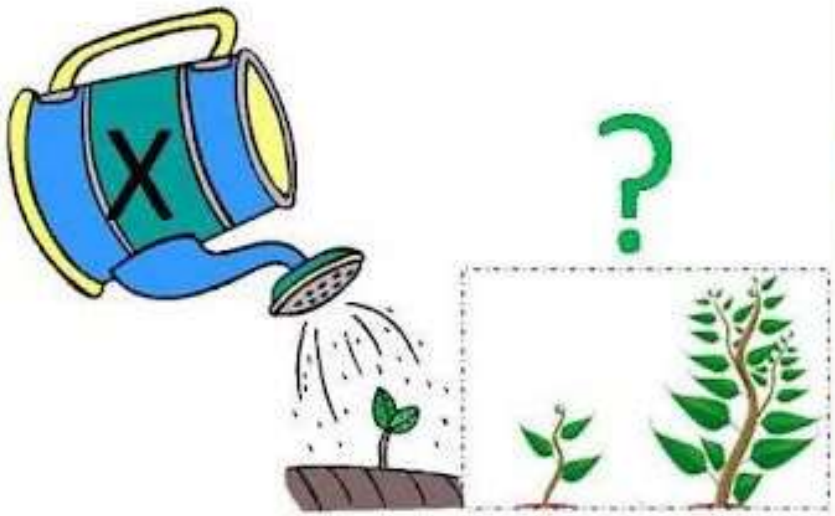
SCIENCENOTES.ORG

Effect of Bio-fertilizer 'x' on Plant growth

www.majordifferences.com

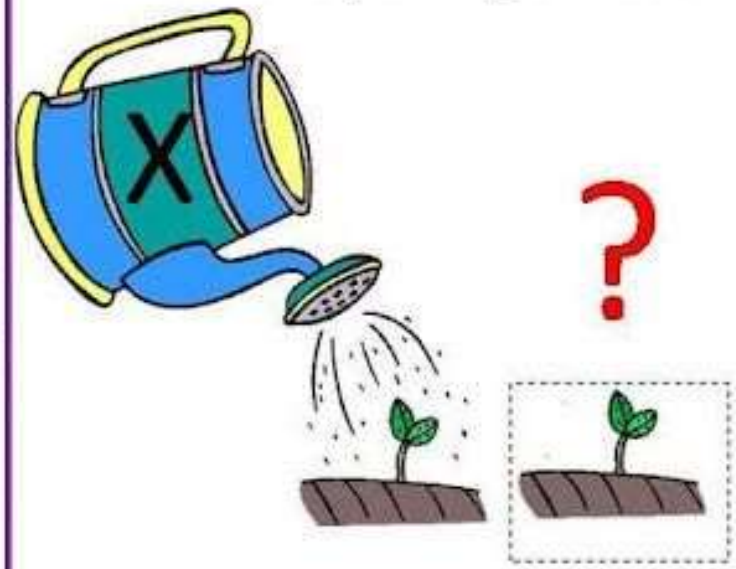
Alternative Hypothesis

H_1: Application of bio-fertilizer 'x' increase plant growth.



Null Hypothesis

H_0: Application of bio-fertilizer 'x' do not increase plant growth.



Determine which variables are relevant to the question.

Determine what kind of variable each one is.

Measurement variables are, as the name implies, things you can measure.

"**Numeric" or "quantitative**" variables.

continuous variables, can have any value in a infinite series For example, length, weight, pH, and bone density.

discrete (or meristic) variables, which only have whole number values; these are things you count, such as the number of spines on an isopod's antenna.

"Nominal variables" are also called categorical, discrete, qualitative, or attribute variables. For example: males and females within a species, categorization of an animal population into juvenile, subadults and adults; red, yellow, white coloured flowers of a plant species.

Do the experiment.

You should do a lot of thinking, planning, and decision-making *before* you do an experiment

You'll have an experiment that is easy to understand, easy to analyze and interpret, answers the questions you're trying to answer, and is neither too big nor too small.

1. Design experimental treatments to manipulate your **independent variable**

2. Assign subjects to groups, either **between-subjects** or **within-subjects**

3. Plan how you will measure your **dependent variable**

For valid conclusions, you also need to select a representative sample and control any extraneous variables that might influence your results. If if random assignment of participants to control and treatment groups is impossible, unethical, or highly difficult, consider an observational study instead.

Based on the number of variables, the kinds of variables, the expected fit to the parametric assumptions, and the hypothesis to be tested, choose the best statistical test to use.

Communicate your results effectively, usually with a graph or table.

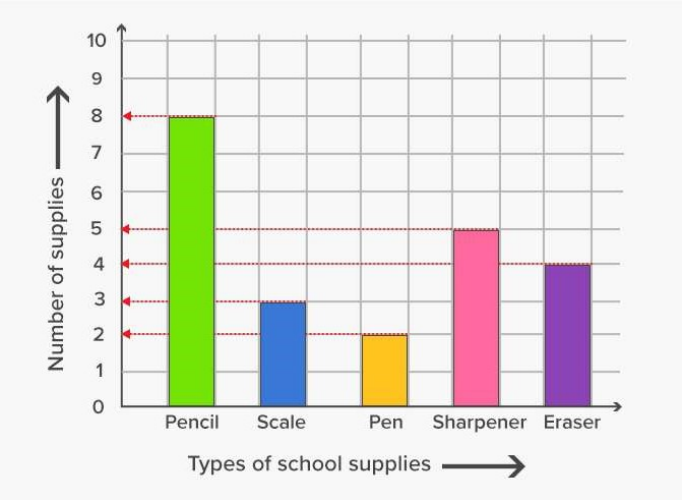
Data tables display information in a grid-like format of rows and columns. They organize information in a way that's easy to scan so that users can look for patterns and develop insights from data.

Table 1			
Week	Total grass height (inches)		
	Trial A	Trial B	Trial C
0	0.50	0.50	0.50
1	1.26	1.01	0.94
2	2.21	1.48	1.13
3	3.52	2.03	1.20

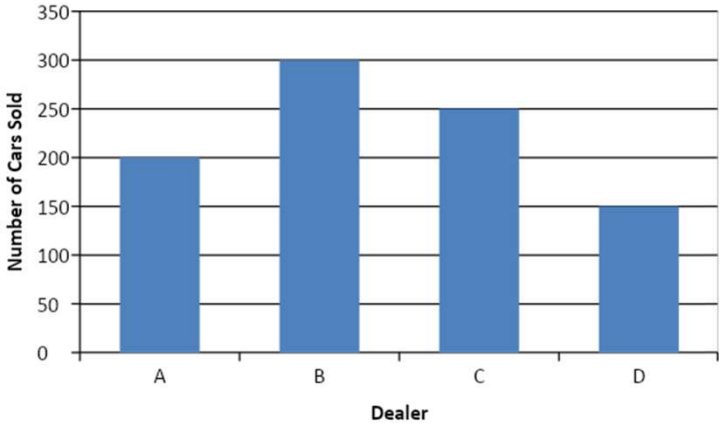
Plot ID	Temperature(°C)	Yield (kg)
A	19	115
B	22	146
C	27	132
D	30	94

a graph can be defined as a pictorial representation or a diagram that represents data or values in an organized manner.

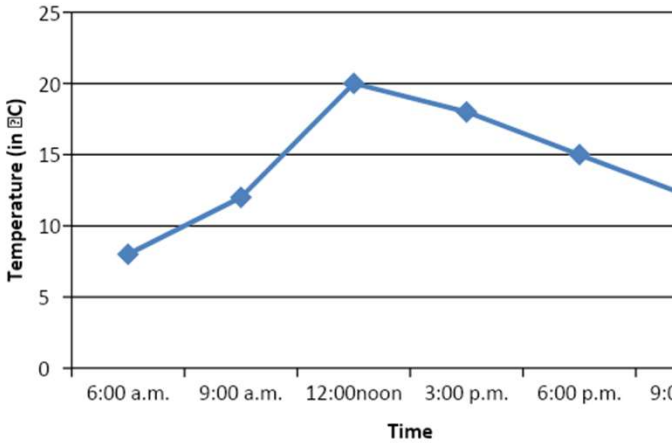
The points on the graph often represent the relationship between two or more things.



Bar Graph

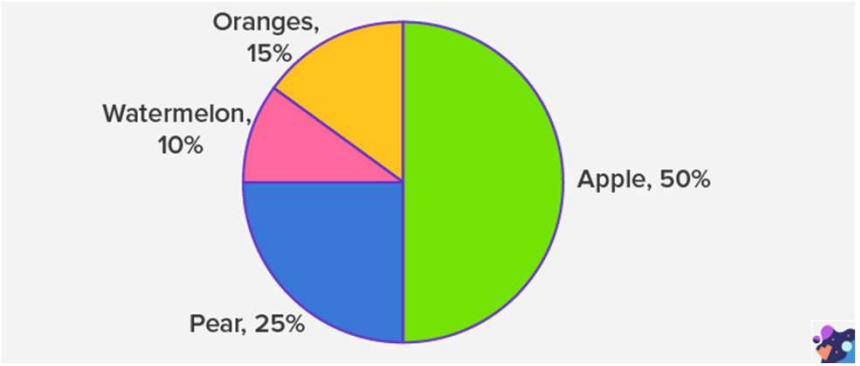


Line Graph:



Pie Chart:

The pie chart is also known as a circle graph. It shows how a whole is divided into different parts. The pie chart shows the relative size of each data set in proportion to the entire data set. Percentages are used to show how much of the whole each category occupies.



LET'S DO A SMALL EXPERIMENT

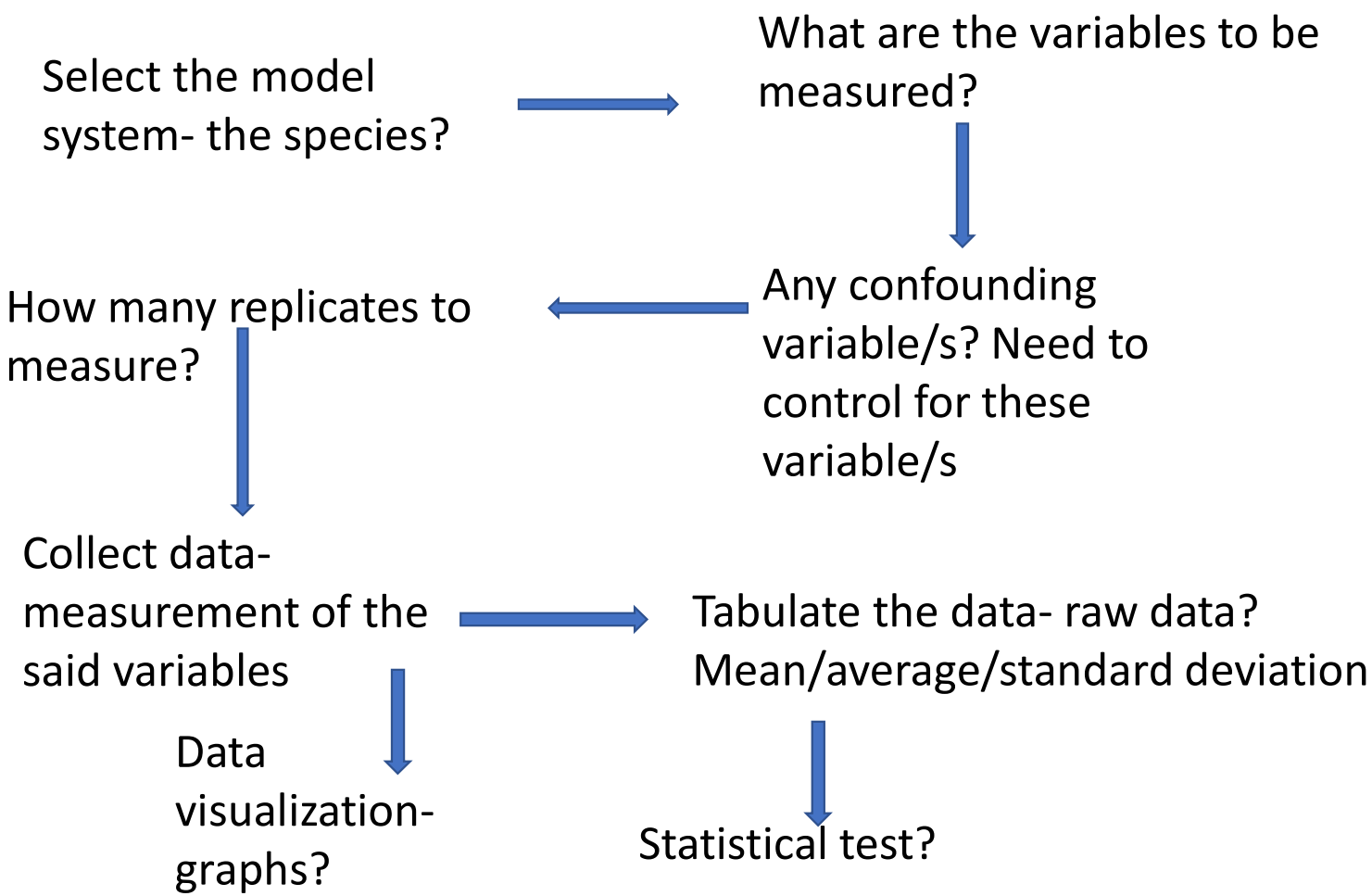
Is there a relationship between plant height, stem girth in any given plant species?

Or

Does exposure to sunlight affect plant height?

What are the null and alternate hypotheses here?

PROTOCOL


Select the model system- the species?
What are the variables to be measured?
Any confounding variable/s? Need to control for these variable/s
How many replicates to measure?
Collect data- measurement of the said variables
Tabulate the data- raw data? Mean/average/standard deviation
Data visualization- graphs?
Statistical test?

Instructions

1. Collection of data can be done in groups of 2-3 students each.

2. Select a species you are familiar with on this campus- choose a shrub, as its most convenient to study.

3. Replicates of 10-15 individual plants (of the same species) should be taken. Measure the plant girth and plant height using a string and then taking its length with a scale.

4. Tabulate your data: Make a table showing individual plant height and its girth. Calculate their mean (average) and standard deviation

5. Show it in the form of graphs: make a bar plot to show this.
6. Make a scatterplot to show relationship between plant height (x axis) and plant girth (y axis).
7. What do you conclude based on the visual assessment of your graph?